RetrievalBench: Chunking and Embedding Choice as First-Class Variables in Cross-Domain Retrieval Evaluation

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Abstract

Standard retrieval benchmarks evaluate systems over pre-segmented corpora, leaving document segmentation outside the experimental design. Production retrieval systems do not have this luxury: documents must be divided into retrievable units, and the segmentation policy may interact with both the corpus and the embedding model. We present **RetrievalBench**, a cross-domain evaluation framework that treats chunking strategy and embedding-model choice as controlled experimental variables. Our main study is a complete sensitivity matrix over twelve domains, four chunking strategies, and three dense embedding models, with BM25 included as a sparse reference and bootstrap confidence intervals computed for every configuration. Chunking is not a neutral preprocessing step: holding the retriever and corpus fixed, its selection changes NDCG@10 by as much as 47% relative, and no chunker is uniformly best. We introduce the Chunking Sensitivity Index (CSI) and Embedding Sensitivity Index (ESI), a per-domain variance decomposition that identifies which decision deserves greater tuning attention. CSI exceeds ESI in 5 of 12 domains, especially in several structured or long-form collections, whereas embedding choice dominates in many short-passage matching tasks. These results replace the claim that "chunking always matters more" with a domain-dependent diagnostic. The framework additionally records retrieval-stage latency and metered embedding-API usage, but we treat these measurements as exploratory rather than universal deployment estimates.

1 Introduction

Retrieval evaluation is commonly built around a fixed corpus of atomic documents, a query set, relevance judgments, and a ranking metric such as NDCG@10 [1]. This abstraction has enabled reproducible comparison across domains, but it hides a consequential decision faced by deployed retrieval systems: source documents must often be segmented before they can be indexed. A financial report can be split into windows, sentences, paragraphs, or semantically coherent groups. Each choice changes the units presented to the retriever and, consequently, the ranking that is evaluated.

Chunking is frequently discussed as an implementation detail. That treatment is questionable for two reasons. First, chunk boundaries determine which evidence is represented by a single vector or sparse document. Second, a chunking rule can interact with the embedding model: two models may respond differently to short sentences, long windows, and mixed-topic passages. Evaluating one fixed segmentation therefore risks attributing a preprocessing effect to the retriever itself.

RetrievalBench makes these decisions explicit. For each domain, we construct four chunked variants of a common document sample and evaluate dense retrieval with three embedding models. We include BM25 as a sparse reference, propagate document-level relevance judgments to chunks by parent-document identity, and aggregate retrieved chunks back to documents for evaluation. The result is not intended as a new absolute leaderboard. Because the main matrix uses controlled subsamples, it supports within-matrix comparisons among chunkers and embedders rather than comparison with full-corpus leaderboard scores.

Our contributions are:

1. A controlled cross-domain benchmark covering twelve domains, four operationally specified chunking strategies, three dense embedding models, and a BM25 reference.

2. Empirical evidence that chunking choice changes NDCG@10 by up to 47% relative and interacts with both domain and retrieval model; no single strategy is uniformly best.
3. CSI and ESI, variance-based diagnostics that compare chunking sensitivity with embedding-model sensitivity within each domain. The analysis shows that their relative importance is domain-dependent rather than universal.
4. An artifact-oriented evaluation implementation with bootstrap uncertainty estimates, paired-comparison utilities, metered OpenAI embedding usage, and optional latency and reranking analyses. We distinguish these supported capabilities from the central matrix claims.

2 Related Work

Cross-domain retrieval evaluation. BEIR standardized heterogeneous zero-shot retrieval evaluation across biomedical, scientific, financial, fact-checking, question-answering, and other domains [1]. Reproducible IR toolkits have similarly made sparse and dense baselines easier to compare [2]. RetrievalBench retains standard document-level metrics but varies the segmentation under which a document is indexed.

Dense retrieval and embedding models. Dense passage retrieval represents queries and candidate text in a shared vector space [3]. General-purpose embedding benchmarks broadened evaluation across tasks [4], while model families such as E5 and BGE provide strong compact retrievers [5, 6]. Our focus is not a new embedding architecture, but the interaction between the embedder and the chunked units it consumes.

Chunking and reranking. Fixed windows, sentence grouping, recursive splitting, and semantic grouping are widely available in RAG tooling, yet they are rarely crossed systematically with both domain and embedding choice. Cross-encoder reranking is a standard second stage [7]. RetrievalBench supports reranking, but the central sensitivity matrix deliberately excludes reranked arms so that CSI and ESI measure chunking and embedding effects without introducing a second-stage model.

3 Experimental Framework

3.1 Domains and sampling

The matrix contains twelve public-data domains: finance (FiQA), legal question answering (Open Australian Legal QA), biomedical retrieval (TREC-COVID), medical retrieval (NFCorpus), scientific claim verification (SciFact), technical literature retrieval (SciDocs), argumentation (ArguAna), community duplicate-question retrieval (Quora), open-domain question answering (Natural Questions), entity retrieval (DBpedia-Entity), fact checking (FEVER), and multi-hop question answering (HotpotQA).

For comparability and tractable repeated indexing, we apply a seeded sampling policy. Each corpus contains 2,000 sampled fill documents plus every document referenced by the retained relevance judgments. Consequently, a corpus can exceed 2,000 documents when the relevant set alone is larger. We retain at most 500 seeded-sampled queries, except where the dataset contains fewer eligible queries or where the legal collection is evaluated in full. Sampling uses seed 42. Table 1 reports the realized sizes.

Subsampling reduces the number of distractors and can inflate absolute effectiveness. We therefore use the matrix to compare configurations evaluated on the same sampled corpus; we do not present its absolute scores as BEIR leaderboard replacements.

3.2 Operational chunking strategies

We evaluate four strategies, described here according to the implementation used to produce the reported matrix:

- **Fixed-512**: non-overlapping progress through whitespace-delimited text using windows of at most 512 words and an overlap of 50 words.
- **Sentence**: non-overlapping windows of three sentences, using NLTK sentence segmentation.
- **Recursive**: split first on blank-line paragraph boundaries; paragraphs exceeding 512 whitespace-delimited words are split into sentences.

Table 1: Domains and realized matrix sizes. Corpus subsampling always retains qrels-relevant documents, so some collections exceed 2,000 documents.

Domain	Dataset	Documents	Queries
Finance	fiqa	2,000	500
Legal	auslegalqa	2,099	2,124
Biomedical	trec-covid	17,537	50
Argumentation	arguana	1,999	500
Open-domain QA	nq	2,000	500
Medical	nfcorpus	3,128	323
Scientific	scifact	2,000	300
Technical	scidocs	2,180	500
Community	quora	2,000	500
Fact checking	fever	2,000	500
Multi-hop QA	hotpotqa	2,000	500
Entity retrieval	dbpedia-entity	14,877	400

- **Semantic**: embed sentences with OpenAI `text-embedding-3-small` and greedily merge adjacent sentences when cosine similarity is at least 0.75. These boundaries are computed independently of the downstream dense retriever and reused across embedding-model comparisons.

These names are operational labels rather than claims that the implementations exhaust all possible fixed, sentence-based, recursive, or semantic chunkers. This distinction is important: our conclusions concern sensitivity to these four concrete policies.

3.3 Retrievers and embedders

The sparse reference is BM25. Dense retrieval uses OpenAI `text-embedding-3-small` (1,536 dimensions), `BAAI/bge-small-en-v1.5` (384 dimensions), and `intfloat/e5-small-v2` (384 dimensions). BGE queries use the model's retrieval instruction; E5 queries and passages use the `query:` and `passage:` prefixes. Embeddings are L2-normalized and scored by inner product.

For each query, the retriever selects the top 50 chunks. Chunk scores are aggregated to their parent documents by maximum score, and the top 10 unique documents are evaluated. Relevance is inherited from document-level qrels: a retrieved chunk is treated as evidence for its parent document, but no claim is made that every chunk of a relevant document is independently relevant.

3.4 Metrics and uncertainty

We report NDCG@10 as the primary metric, with Recall@10, MRR, and MAP available in the result artifacts. For each configuration, we compute a 95% bootstrap confidence interval over per-query NDCG using 1,000 seeded resamples. The framework also implements paired permutation tests and Cohen's d for analyses where aligned per-query values are retained.

3.5 Chunking and embedding sensitivity

For a fixed domain, let $M_{c,e}$ denote NDCG@10 for dense retrieval with chunker c and embedder e. Let C and E be the sets of four chunkers and three embedders. We define

$$\mathrm{CSI} = \frac{\frac{1}{|E|}\sum_{e\in E} \mathrm{Var}_{c\in C}(M_{c,e})}{\mathrm{Var}_{c\in C, e\in E}(M_{c,e})}, \tag{1}$$

$$\mathrm{ESI} = \frac{\frac{1}{|C|}\sum_{c\in C} \mathrm{Var}_{e\in E}(M_{c,e})}{\mathrm{Var}_{c\in C, e\in E}(M_{c,e})}. \tag{2}$$

CSI $>$ ESI means that, within the evaluated pool, changing chunkers moves effectiveness more than changing embedders on average. The indices are diagnostics, not percentages of variance in a fully orthogonal ANOVA: interaction can cause CSI and ESI not to sum to one.

4 RESULTS

4.1 CHUNKING CHANGES RETRIEVAL EFFECTIVENESS

Table 2 reports the complete matrix. Chunking causes substantial within-system variation. The largest observed relative swing is for biomedical BM25, from 0.476 with semantic chunking to 0.700 with fixed-512: $(0.700 - 0.476)/0.476 = 47.1\%$. Finance shows a similarly consistent pattern across all three dense models: fixed-512 exceeds semantic by 26% relative for OpenAI, 40% for BGE, and 41% for E5.

The best chunker is neither global nor model-independent. Recursive and fixed windows are often competitive on long or structured text, while sentence or semantic policies sometimes win in other domains. Quora supplies a useful boundary condition: its documents are already short and nearly atomic, so all four policies produce almost identical scores. This supports a qualified conclusion: chunking sensitivity depends strongly on the units present in the source corpus.

4.2 CHUNKING VERSUS EMBEDDING IS DOMAIN-DEPENDENT

Table 3 summarizes the variance decomposition. CSI exceeds ESI in 5 of 12 domains: TREC-COVID, FiQA, ArguAna, Natural Questions, and the legal collection. Embedding choice dominates strongly in NFCorpus, SciDocs, HotpotQA, SciFact, Quora, and FEVER. DBpedia-Entity is relatively balanced but remains embedding-dominant under the strict comparison.

These results reject both simple extremes. Chunking is not merely preprocessing, because it can move quality substantially while the retriever is fixed. But it is also not universally more important than the embedding model. A practical reading is to estimate both sensitivities on a representative sample: prioritize chunking experiments when CSI is high, model selection when ESI is high, and joint tuning when the values are close.

The result is conditional on the evaluated model pool. The two local embedders are similarly sized small models, whereas the API model belongs to a different capacity and training regime. This may increase observed embedding variance. Conversely, the four chunkers represent only one parameterization of each strategy family. CSI and ESI should therefore be interpreted as properties of a domain under a declared candidate set, not immutable corpus constants.

4.3 EXPLORATORY DEPLOYMENT MEASUREMENTS

RetrievalBench records metered OpenAI token usage and supports a latency-discounted score,

$$\text{LA-NDCG@10} = \text{NDCG@10} \exp\left(-\frac{\max(0, \ell - B)}{B}\right), \tag{3}$$

where B is a user-selected latency budget. We include these facilities because deployment choices depend on more than effectiveness. However, the archived latency runs were collected on a single machine and do not share a fully production-complete online boundary: offline indexing is excluded, and dense timing does not include all API/network overhead. We therefore do not use them to claim a universal system reordering. They are examples of how the framework can be applied once a deployment defines its own measurement boundary.

The reported cost field is similarly narrow: it is metered OpenAI embedding API spend at the price used during the experiments. It includes OpenAI calls used for semantic segmentation and OpenAI dense embeddings, but assigns no monetary value to local-model compute, hardware, energy, or reranker inference. It should not be interpreted as total serving cost.

5 DISCUSSION

Chunking is part of the evaluated system. If two studies evaluate the same embedder using different segmentation policies, their scores need not estimate the same system. Benchmark reports should therefore publish chunker code and parameters alongside model identifiers and retrieval depth.

Short-document domains are an informative control. The near-zero spread on Quora shows that a benchmark need not reward arbitrary chunker complexity. When documents are already atomic, segmentation has little room to change the indexed unit. This also cautions against extrapolating chunking conclusions from passage collections to long documents.

Sensitivity indices guide experimental allocation. CSI and ESI do not nominate a universal winner. Their value is operational: a small pilot matrix can show whether engineering time is better spent exploring segmentation, models, or their interaction.

6 LIMITATIONS

First, the main matrix uses one seeded corpus/query sample per domain. Although every configuration within a domain shares that sample, additional seeds would better quantify sensitivity to distractor selection. Second, propagating document-level qrels to parent-linked chunks does not identify which specific chunk contains the judged evidence; chunk-level annotation would support a stricter analysis. Third, absolute matrix scores are inflated by subsampling and must not be compared directly with full-corpus leaderboards.

Fourth, the embedder pool mixes an API model with two similarly sized open-source models; a same-capacity model pool would be a useful robustness check. Fifth, the four chunkers are concrete, relatively simple implementations. Different tokenizers, sentence windows, recursive separators, semantic thresholds, and overlap values could change CSI. Sixth, semantic boundaries are generated with the OpenAI model for all downstream embedders; this controls the segmentation across systems but may favor semantic judgments aligned with that model.

Finally, latency, reranking, and monetary cost are supported by the framework but are not established here as universal cross-domain conclusions. Existing latency runs use a single environment and a limited timing boundary, local compute is not monetized, and some archived reranker runs produced unchanged rankings on out-of-domain tasks. We retain these analyses as exploratory artifacts rather than treating them as central evidence.

7 CONCLUSION

RetrievalBench evaluates a decision that fixed-corpus benchmarks usually hide. Across twelve domains, changing only the chunking strategy changes NDCG@10 by as much as 47% relative, and the preferred strategy varies across domains and retrievers. CSI and ESI further show that the chunking-versus-embedding question has no universal answer: chunking dominates in 5 of 12 domains, while embedding choice dominates in the remainder of the evaluated pool. The broader lesson is methodological. Segmentation should be reported and evaluated as part of the retrieval system, and its importance should be measured rather than assumed.

ARTIFACT AND REPRODUCIBILITY STATEMENT

The repository contains the per-domain aggregate result files used for Tables 2 and 3, the seeded sampling and experiment runner, bootstrap metric implementation, and CSI/ESI analysis script. The reported matrix was generated with:

```
RB_MAX_CORPUS=2000 RB_MAX_QUERIES=500 \
python scripts/run_chunking_pipeline.py --dataset <dataset> \
  --embedders openai bge-small e5-small
```

Some large-relevant-set collections exceed the nominal corpus cap because all qrels-relevant documents are retained. Aggregate artifacts are available for all twelve domains. Complete per-query ranked-list sidecars are not available for every historical run; accordingly, we do not claim that every published aggregate can be reconstructed from ranked lists alone. A table-generation and artifact-validation script should accompany the archival release to verify domain, chunker, embedder, and value completeness directly from the released JSON files.

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Table 2: NDCG@10 for the complete sensitivity matrix. BM25 is a sparse reference; the three dense columns form the 4×3 matrix used for CSI/ESI. Compare configurations within a domain, not with full-corpus leaderboards.

Domain	Chunker	BM25	Dense–OpenAI	Dense–BGE	Dense–E5
Finance	Sentence	.285	.676	.627	.607
	Fixed-512	.346	.739	.677	.665
	Recursive	.329	.726	.663	.624
	Semantic	.254	.585	.485	.471
Legal	Sentence	.688	.793	.764	.811
	Fixed-512	.732	.816	.771	.824
	Recursive	.732	.816	.771	.824
	Semantic	.659	.765	.666	.764
Biomedical	Sentence	.619	.928	.916	.907
	Fixed-512	.700	.948	.964	.920
	Recursive	.690	.943	.944	.906
	Semantic	.476	.879	.854	.839
Argumentation	Sentence	.416	.571	.557	.492
	Fixed-512	.419	.566	.598	.536
	Recursive	.434	.569	.599	.537
	Semantic	.363	.503	.471	.428
Open-domain QA	Sentence	.816	.980	.978	.975
	Fixed-512	.839	.980	.979	.981
	Recursive	.841	.979	.979	.965
	Semantic	.785	.985	.960	.950
Medical	Sentence	.269	.399	.363	.329
	Fixed-512	.280	.403	.357	.340
	Recursive	.279	.402	.357	.334
	Semantic	.251	.386	.368	.316
Scientific	Sentence	.583	.797	.785	.717
	Fixed-512	.635	.798	.784	.764
	Recursive	.638	.792	.780	.746
	Semantic	.522	.778	.755	.706
Technical	Sentence	.309	.428	.407	.338
	Fixed-512	.327	.439	.434	.392
	Recursive	.328	.434	.432	.344
	Semantic	.281	.412	.394	.326
Community	Sentence	.955	.997	.998	.996
	Fixed-512	.955	.997	.998	.996
	Recursive	.955	.997	.998	.996
	Semantic	.954	.997	.997	.991
Fact checking	Sentence	.954	.986	.984	.989
	Fixed-512	.949	.985	.984	.988
	Recursive	.936	.984	.983	.987
	Semantic	.939	.985	.987	.987
Multi-hop QA	Sentence	.850	.937	.932	.949
	Fixed-512	.856	.942	.931	.949
	Recursive	.856	.942	.931	.949
	Semantic	.823	.938	.929	.937
Entity retrieval	Sentence	.644	.855	.821	.842
	Fixed-512	.651	.860	.823	.865
	Recursive	.651	.860	.824	.865
	Semantic	.618	.846	.825	.792

Table 3: Chunking and embedding sensitivity for pure dense retrieval.

Domain	Dataset	CSI	ESI	CSI>ESI
Biomedical	trec-covid	.858	.184	Yes
Finance	fiqa	.770	.244	Yes
Argumentation	arguana	.723	.338	Yes
Open-domain QA	nq	.710	.677	Yes
Entity retrieval	dbpedia-entity	.643	.717	No
Legal	auslegalqa	.566	.507	Yes
Fact checking	fever	.483	.787	No
Community	quora	.381	.856	No
Scientific	scifact	.292	.800	No
Multi-hop QA	hotpotqa	.223	.874	No
Technical	scidocs	.222	.826	No
Medical	nfcorpus	.059	.985	No